

LAB ACTIVITY (Lab_01-03)

1) Arithmetic Operators

```
import java.util.*;
public class arithmetic
{
    public static void main(String[] args) {
        int num1, num2;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter number 1");
        num1=sc.nextInt();
        System.out.println("Enter number 2");
        num2=sc.nextInt();

        int sum=num1+num2;
        int sub=num1-num2;
        int mul=num1*num2;
        int div=num1/num2;

        System.out.println("Results:");
        System.out.println("Addition:"+ sum);
        System.out.println("Subtraction:"+ sub);
        System.out.println("Multiplication:"+ mul);
        System.out.println("Division:"+ div);
    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>javac arithmetic.java
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>java arithmetic
Enter number 1
25
Enter number 2
5
Results:
Addition:30
Subtraction:20
Multiplication:125
Division:5
```

2) Unary Operators

```
import java.util.*;
public class unary
{
    public static void main(String[] args) {
        int num;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter number 1");
        num=sc.nextInt();

        System.out.println("Results:");
        System.out.println("Pre-Increment:"+ ++num);
        System.out.println("Post-Increment:"+ num++);
        System.out.println("Post-Decrement:"+ num-- );
        System.out.println("Pre-Decrement:"+ --num);
    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>javac unary.java
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>java unary
Enter number 1
23
Results:
Pre-Increment:24
Post-Increment:24
Post-Decrement:25
Pre-Decrement:23
```

3) Relational Operators

```
import java.util.*;
public class relational
{
    public static void main(String[] args) {
        int num1, num2;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter number 1");
        num1=sc.nextInt();
        System.out.println("Enter number 2");
        num2=sc.nextInt();

        System.out.println("Results:");
        if(num1>num2){
            System.out.println("Number 1 is greater than Number 2");}
        else if(num2>num1){
            System.out.println("Number 2 is greater than Number 1");}
        else{
            System.out.println("Number 1 and Number 2 are equal");}

    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>javac relational.java
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>java relational
Enter number 1
23
Enter number 2
56
Results:
Number 2 is greater than Number 1
```

4) Logical Operators

```
import java.util.*;
public class logical
{
    public static void main(String[] args) {
        int a, b;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter number 1");
        a=sc.nextInt();
        System.out.println("Enter number 2");
        b=sc.nextInt();

        boolean and= (a>0) && (b>0);
        boolean or= (a>0) || (b>0);
        boolean not=!(a>b);

        System.out.println("Results:");
        System.out.println("(a>0) && (b>0):"+ and);
        System.out.println("(a>0) || (b>0):"+ or);
        System.out.println("!(a>b):"+ not);

    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>javac logical.java
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>java logical
Enter number 1
23
Enter number 2
-4
Results:
(a>0) && (b>0):false
(a>0) || (b>0):true
!(a>b):false
```

5) Bitwise Operators

```
import java.util.*;
public class bitwise
{
    public static void main(String[] args) {
        int a, b;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter number 1");
        a=sc.nextInt();
        System.out.println("Enter number 2");
        b=sc.nextInt();

        int and= a&b;
        int or= a|b;
        int not= ~a;
        int ls=a<<1;
        int rs=a>>1;

        System.out.println("Results:");
        System.out.println("a&b:"+ and);
        System.out.println("a|b:"+ or);
        System.out.println("~a:"+ not);
        System.out.println("a<<1:"+ ls);
        System.out.println("a>>1:"+ rs);

    }
}
```

```
D:\SEMESTER I\LAB WORK\Java Programming\lab2(2308)>java bitwise
Enter number 1
5
Enter number 2
3
Results:
a&b:1
a|b:7
~a:-6
a<<1:10
a>>1:2
```

